

Week 3 Technical Report: Statistics and Probability Analysis

AnalystLab Africa Data Science Internship

Introduction

This report presents the statistical analysis completed during Week 3 of the AnalystLab Africa Data Science Internship. The objective of this week's activities was to apply statistical concepts to the cleaned Titanic and Housing datasets developed during Weeks 1&2. The analysis focused on descriptive statistics, probability distributions, hypothesis testing, and understanding the distinction between correlation and causation.

These techniques provide valuable insights into the characteristics of the data and support evidence-based decision-making in data science.

Objectives

The objectives of this analysis were to:

- Calculate descriptive statistics for both datasets.
- Explore probability distributions of key numerical variables.
- Perform hypothesis testing to investigate relationships between variables.
- Differentiate between correlation and causation using practical examples.
- Interpret statistical findings to support data-driven conclusions.

Descriptive Statistics

Descriptive statistics were calculated for all numerical variables in both datasets, including the mean, median, mode, minimum, maximum, variance, standard deviation, quartiles, and interquartile range.

For the Titanic dataset, the statistics provided insights into passenger characteristics such as age and fare distributions. The Housing dataset revealed the variation in property prices, house sizes, and other numerical features.

DESCRIPTIVE STATISTICS - TITANIC DATASET											
	Mean	Median	Mode	Minimum	Maximum	Range	Variance	Standard Deviation	Q1	Q3	IQR
PassengerId	446.000000	446.0000	1.00	1.00	891.0000	890.0000	66231.000000	257.353842	223.5000	668.5	445.0000
Survived	0.383838	0.0000	0.00	0.00	1.0000	1.0000	0.236772	0.486592	0.0000	1.0	1.0000
Pclass	2.308642	3.0000	3.00	1.00	3.0000	2.0000	0.699015	0.836071	2.0000	3.0	1.0000
Age	29.361582	28.0000	28.00	0.42	80.0000	79.5800	169.512498	13.019697	22.0000	35.0	13.0000
SibSp	0.523008	0.0000	0.00	0.00	8.0000	8.0000	1.216043	1.102743	0.0000	1.0	1.0000
Parch	0.381594	0.0000	0.00	0.00	6.0000	6.0000	0.649728	0.806057	0.0000	0.0	0.0000
Fare	32.204208	14.4542	8.05	0.00	512.3292	512.3292	2469.436846	49.693429	7.9104	31.0	23.0896


```

# Descriptive statistics for the housing dataset
print("DESCRIPTIVE STATISTICS - HOUSING DATASET")
display(descriptive_statistics(housing))
✓ 0.1s

```

DESCRIPTIVE STATISTICS - HOUSING DATASET											
	Mean	Median	Mode	Minimum	Maximum	Range	Variance	Standard Deviation	Q1	Q3	IQR
price	4.766729e+06	4340000.0	3500000.0	1750000.0	13300000.0	11550000.0	3.498544e+12	1.870440e+06	3430000.0	5740000.0	2310000.0
area	5.150541e+03	4600.0	6000.0	1650.0	16200.0	14550.0	4.709512e+06	2.170141e+03	3600.0	6360.0	2760.0
bedrooms	2.965138e+00	3.0	3.0	1.0	6.0	5.0	5.447383e-01	7.380639e-01	2.0	3.0	1.0
bathrooms	1.286239e+00	1.0	1.0	1.0	4.0	3.0	2.524757e-01	5.024696e-01	1.0	2.0	1.0
stories	1.805505e+00	2.0	2.0	1.0	4.0	3.0	7.525432e-01	8.674925e-01	1.0	2.0	1.0
parking	6.935780e-01	0.0	0.0	0.0	3.0	3.0	7.423300e-01	8.615858e-01	0.0	1.0	1.0

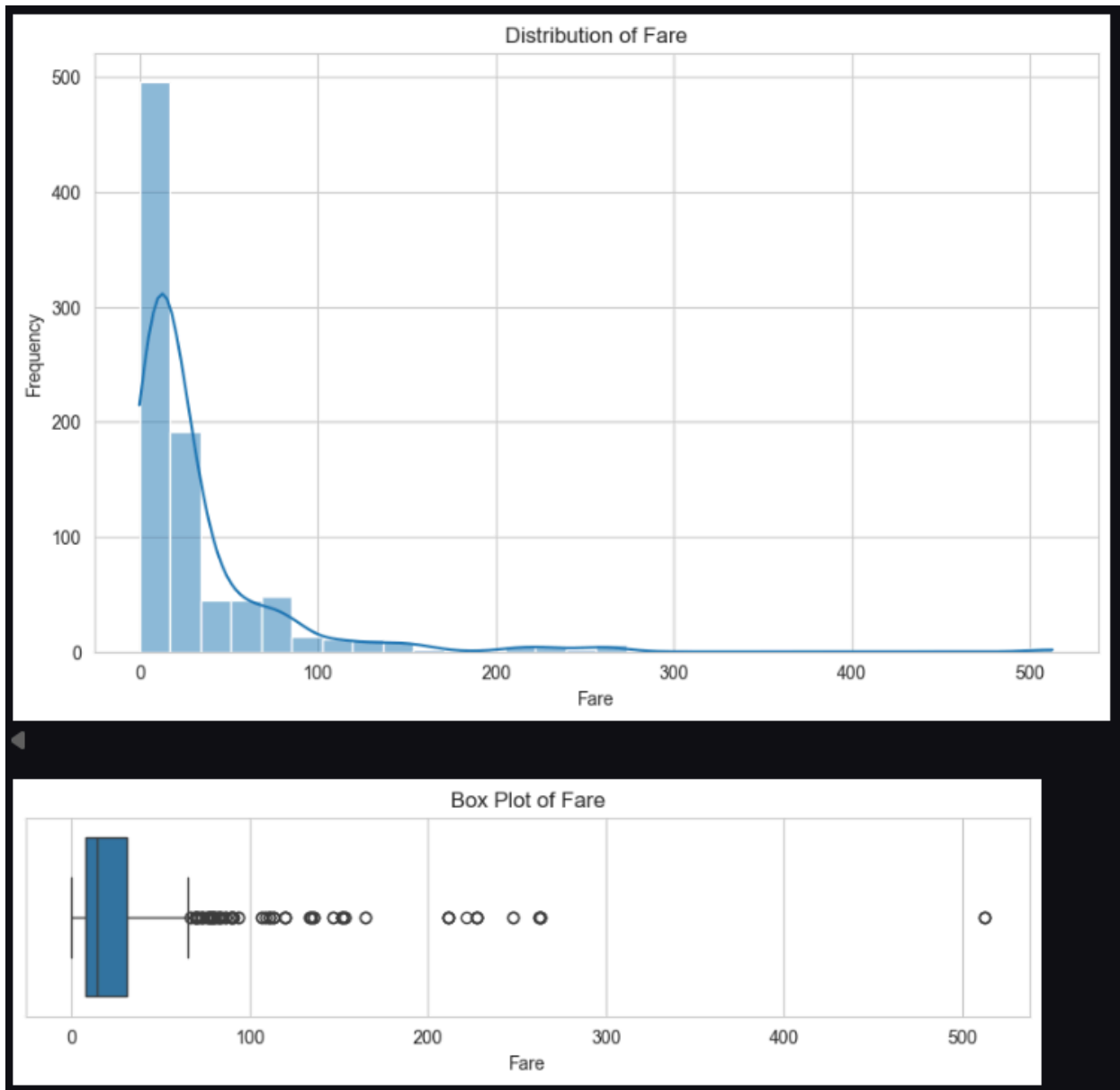
Figure 1: Descriptive statistics for the both the titanic and housing datasets.

These statistics helped identify the central tendency and variability of each variable before conducting further statistical analyses

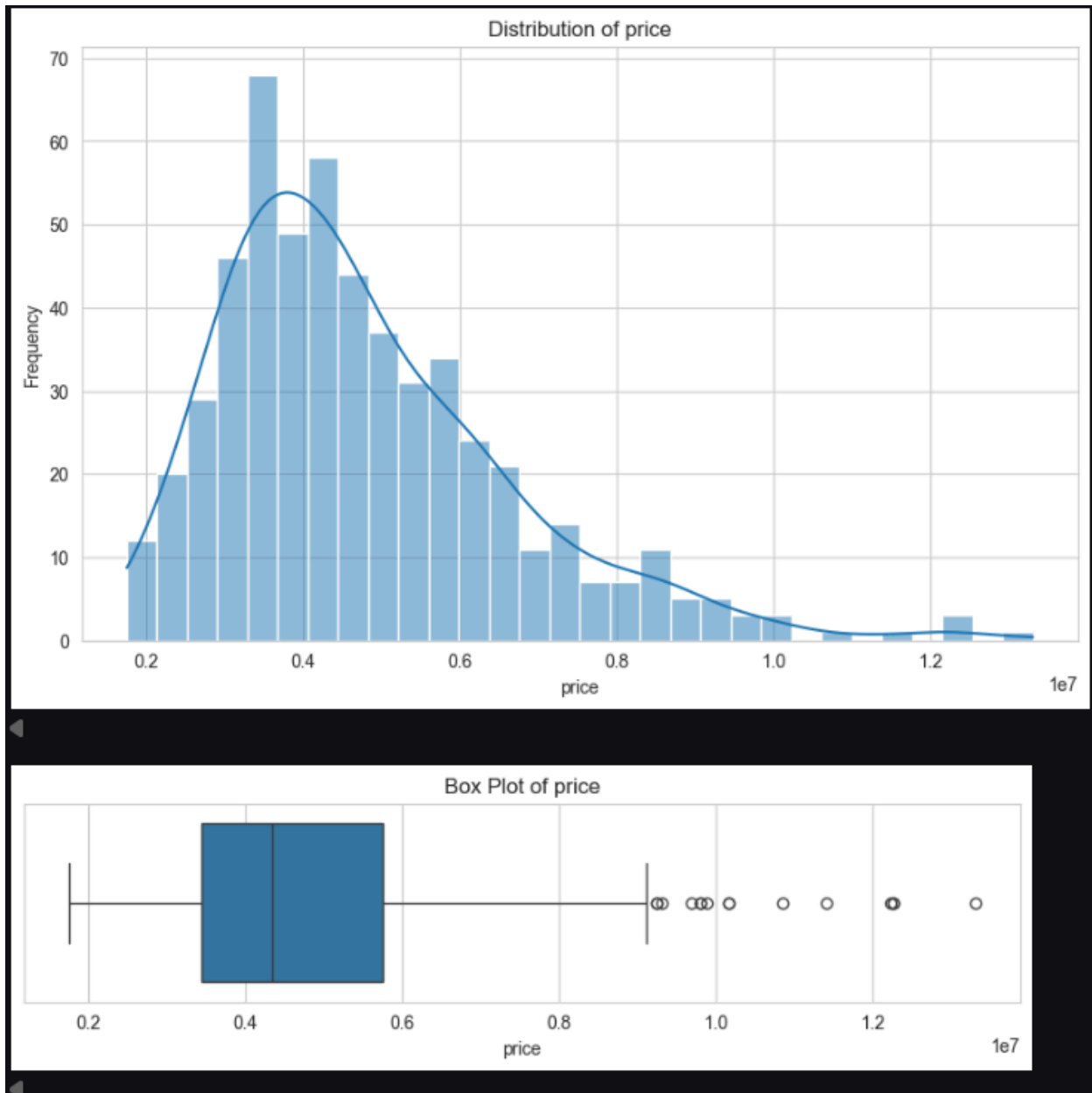
Probability Distributions

Histograms, kernel density estimation (KDE) plots, and box plots were used to examine the distributions of selected numerical variables.

For the Titanic dataset, passenger age and fare distributions were analysed. The fare distribution displayed positive skewness, indicating that a relatively small number of passengers paid substantially higher ticket prices.



For the Housing dataset, house prices and property areas also exhibited positive skewness, reflecting the presence of higher-value properties within the dataset.



The box plots assisted in identifying potential outliers, while the skewness values provided numerical evidence regarding the shape of each distribution.

Hypothesis Testing

Several hypothesis tests were conducted to investigate relationships within the datasets.

For the Titanic dataset:

- An Independent Samples t-test compared the average ages of survivors and non-survivors.
- A Chi-Square Test evaluated the relationship between passenger class and survival.

- A One-Way ANOVA examined whether the average age differed among first-, second-, and third-class passengers.
- A Chi-Square Test investigated whether passenger class influenced female survival rates.

For the Housing dataset:

- A One-Way ANOVA assessed whether the number of bedrooms affected average house prices.
- A Pearson correlation test evaluated the relationship between property area and selling price.
- An Independent Samples t-test compared house prices between preferred and non-preferred residential areas.

The statistical decisions were based on a significance level of 5% ($\alpha = 0.05$), allowing conclusions to be drawn using p-values obtained from each test.

Correlation vs Causation

Correlation analysis identified relationships between variables in both datasets. In the Titanic dataset, survival was associated with passenger class and gender, while in the Housing dataset, house area, bathrooms, bedrooms, and parking spaces showed positive relationships with selling price.

However, these findings do not imply causation. Although variables may move together, additional evidence is required before concluding that one variable directly causes changes in another.

Understanding this distinction is essential for accurate interpretation of statistical results and responsible data analysis.

Key Findings

The statistical analysis produced several important insights:

- Female passengers were more likely to survive than male passengers.
- Passenger class was significantly associated with survival outcomes.
- The average age differed across passenger classes.
- Larger houses generally sold at higher prices.
- Houses with more bedrooms tended to have higher average selling prices.
- House area showed a strong positive relationship with selling price.
- Both datasets contained positively skewed variables and several potential outliers.

Conclusion

Week 3 strengthened my understanding of statistical analysis and its application in data science. By applying descriptive statistics, probability distributions, hypothesis testing, and correlation analysis, I gained practical experience in extracting meaningful insights from real-world datasets.

These statistical techniques provide an important foundation for future topics such as predictive modelling, machine learning, and data-driven decision-making.

References

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